

Bot 2108 Long Question

1. Explain the DNA polymerases enzymes in prokaryotic and eukaryotic enzymes.
 - Both prokaryotic and eukaryotic cells possess multiple DNA polymerases with different designation.
 - The first DNA polymerase is involved in DNA repair rather than DNA replication.
 - The second DNA polymerase is also involved in DNA repair while the third DNA polymerase is the main replicating enzyme.
 - The third DNA polymerase is a multi-subunit enzyme that function as a dimer of these multiple subunits. DNA polymerase enzyme can possess three distinct enzymatic activities.
 - All **DNA polymerases** direct the synthesis of DNA fragments in a 5' to 3' direction.
 - Many DNA polymerases also possess 3' to 5' exonuclease activity. This allows the enzyme to remove nucleotides at the 3-end of the newly synthesized chain.
 - This is often considered a "**proof-reading activity**", which can be used to excise incorrectly incorporated nucleotides before their replacement with the correct bases.
 - Some DNA polymerase also has a 5' to 3' exonuclease activity. This allows the enzyme to remove sequences that have already been synthesized and is involved in joining up discontinuous DNA fragments produced on the lagging strand during DNA replication.
 - DNA polymerases are not able to initiate DNA synthesis on their own. They have an absolute requirement for a free 3' - end onto which the enzyme can add new nucleotides.
 - Thus, the two primers are required to initiate DNA replication in each direction from the origin. DNA synthesis only proceeds in a 5' to 3' direction.
 - Both strands of the DNA molecule must be replicated, but only one of the strands is orientated in a 5' to 3' direction from the origin of replication.
 - This strand is called the leading strand, can be copied in a continue manner and can occur without stopping as a DNA polymerase working in its favor orientation.
 - Replication of the other strand, the **lagging strand**, would appear to have in the 3' to 5' direction, which is not possible.
 - Thus, it replicates in a discontinuous fashion and produces a series of short DNA segment in a 5' to 3' direction (*this was stated by Okazaki et al., 1968*).
 - These are then ligated together to produce the intact newly synthesized strand. The short DNA fragment that produced in the lagged strand is called **Okazaki's fragment**.

Bot 2108 Short Questions

5 marks

1. Give a short note on chromosome.
 - The organization of cellular genetic material exists as a compact mass is confined to a limited volume called *chromosome*.
 - Various activities, such as replication and transcription, forming their new identical material, then their identical must be accomplished within the space of the original cell.
 - Cytogenetics must be defined as the study of the inherited materials or genetic materials that are in the cytoplasm of the cell.
 - It comes from the word **Cyto-** (from **kytos** = **hollow space** (Greek word) combined with the words **Genetics** and become **Cyto-genetics**. *Thus, Cytogenetics means the study of Genetics that contain in the cell.*
2. Briefly describe the viral DNA.
 - Viruses are also known as half dead organisms. When they are alone and living free, they have no functions, metabolic activities and are resting or like a dead organism.
 - When they entered the host cells (like bacteria, human animals, etc.), using the host's materials their genetic material starts to function for production.
 - The genetic materials of some viruses consist of relatively small, single stranded DNA molecules. Once inside a cell the single strand serves as a **template** for making a second strand.
 - This results in double-stranded DNA, then govern the production of more virus particles. Bacteriophages **ΦX174** and **M13** carrying single- stranded DNA while **parvoviruses** which are associated with fetal death and spontaneous abortion in humans have single stranded DNA in the form of covalently closed circle linear DNA.
 - There is also a bacterium that consist single stranded DNA without forming a ring shape and their two ends of DNA strands are in loose position.
3. Mention the function of ribosome.
 - The small 30S subunit initially binds to mRNA. The larger 50S subunit contributes which catalyzes formation of the peptide bonds joining adjacent amino acids.
 - Both subunits contribute to three distinct tRNA binding areas *peptidyl transferase enzyme* known as the *Aminoacyl or A site, the peptidyl or P site, and the exit or E site*.
 - Finally, other regions of the ribosome distributed over the two subunits serve as points of contact for some of the additional proteins that play a role in translation.
 - Surprisingly no proteins are found close to the region between the two tRNAs where peptide bonds are formed.
 - This finding supports the conclusion of biochemical experiments that *peptidyl transferase* is actually a function of the 50S subunit's rRNA rather than any protein component of the ribosome; in other words, the rRNA acts as a *ribozyme (ribosome with catalytic activity)* that joins amino acids together.